Tropicana

SP26-MARK-40150-SS-2X Pricing Analytics

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1 Managerial Section

Debu, I took a look at the historical data for all four zones for 2016-2018, and then used a predictive model to gauge what Tropicana's 2018 profits might have been had Jewel incorporated more frequent and deeper price promotions. Based on my analysis, I recommend that Jewel offer more frequent but shallower price promotions with merchandising.

Although the High and Medium zones had significantly higher profits and units sold than the Low and Cubfighter zones, all four zones followed relatively trends. In 2016, Tropicana 16 oz orange juice (Tr-12) was priced relatively highly, with units sold steadily increasing. Near the end of 2016, however, units sold plummeted, resulting in negative profits which drove Jewel to cut prices down significantly to recapture units sold. Base prices were kept relatively low, with frequent price promotions in all four zones cutting the price even further down to just $0.11 above the wholesale cost ($0.99 retail price, $0.88 wholesale price). While these low prices had the intended effect of restoring units sold even beyond their 2016 levels, they were so low that Jewel barely profited from each sale, earning an actual profit of $139,237 in 2017 compared with $233,075 in 2016. With such lower profits in 2017, Jewel probably attempted to reverse the 2017 strategy for 2018, offering less frequent and shallower price promotions. While this brought units sold well below 2017 levels, they were able to capture enough profit per sale to bring revenue back up to $214,406.

I then ran the linear regression model to predict whether an alternate 2018 scenario (scenario 1) involving more frequent and deeper price promotions would have generated higher profits for Tropicana. The predictive model estimated that running more frequent and shallow promotions would have generated less revenue than Jewel had in 2018, and even 2017. The key to raising Jewel profits is increasing profit margins and using consumer price sensitivity to determine the depth of price promotions. Tropicana performed significantly better in 2018 than in 2017 because even though it sold significantly less units, Jewel was able to maximize profit margins. Scenario 1 produced such underwhelming results because it cut into profit margins while failing to compensate with units sold. At least in 2017, price promotions cut so deep that units sold spiked as a result of those price promotions.

Jewel needs to offer more price promotions because the historical data shows a clear relationship between price decreasing and units sold increasing. Units sold and profits spike each time price promotions are offered, indicating that Jewel should offer more frequent price promotions for Tropicana. How deep should our price promotions be? Here is where the scenario analysis comes in. Jewel's promotion strategy in 2018 successfully recaptured revenue because it incorporated price sensitivity well. The regression model shows that generally speaking, consumers are only slightly price sensitive, but tend to become significantly more price sensitive with merchandising. Jewel can take advantage of this consumer behavior trend by offering more frequent, but shallower price promotions. Continuing merchandise during price promotions will allow Tropicana to capture more customers with only a slight price decrease. This, in turn, will keep profit margins high while generating the spikes in units sold that Tr-12's price promotions have generated in the past.

I would also like to note a few limitations of this case study. The first is that the data only indicates whether there was merchandising or not, giving no information about what merchandising tactics were used. A deeper dive into effective forms of merchandising would allow Jewel to capture more Tr-12 customers without cutting too much into profit margins. Additionally, the data does not provide any clear insights as to why units sold plummeted toward the end of 2016. Because 2016 was Tr-12's most successful year within the scope of this study, diving deeper into Tr-12's 2016 trends may provide further insights into tactics that can sell many units while keeping prices on the higher end. Finally, scenario 1 changes both promotion frequency and depth, obscuring the individual effects that each one has on the scenario 1 profit estimate. I would like to run a similar scenario test that evaluates scenario 0 against a scenario that increases promotion frequency without increasing scenario 0's promotion depth. This will allow our team to better assess whether the increased promotion frequency and smaller promotion depth model will increase Jewel's Tropicana profits.

2 Technical Section

This section contains an analysis of the case study examining historical Tropicana 12-ounce orange juice for 2016 - 2018, as well as a predictive model contrasting 2018 data with data from a proposed alternative scenario (scenario 1), featuring more and deeper price promotions for that year.

```
# Read the data
df <- read_excel("tropicana.xlsx")

# Create some variables
df <- df %>% mutate(ln_p=log(price_tr_12),
                    ln_q=log(units_tr_12),
                    Dmerch=factor(merch_tr_12),
                    Dzone=factor(zone),
                    Dyear=factor(year))
```

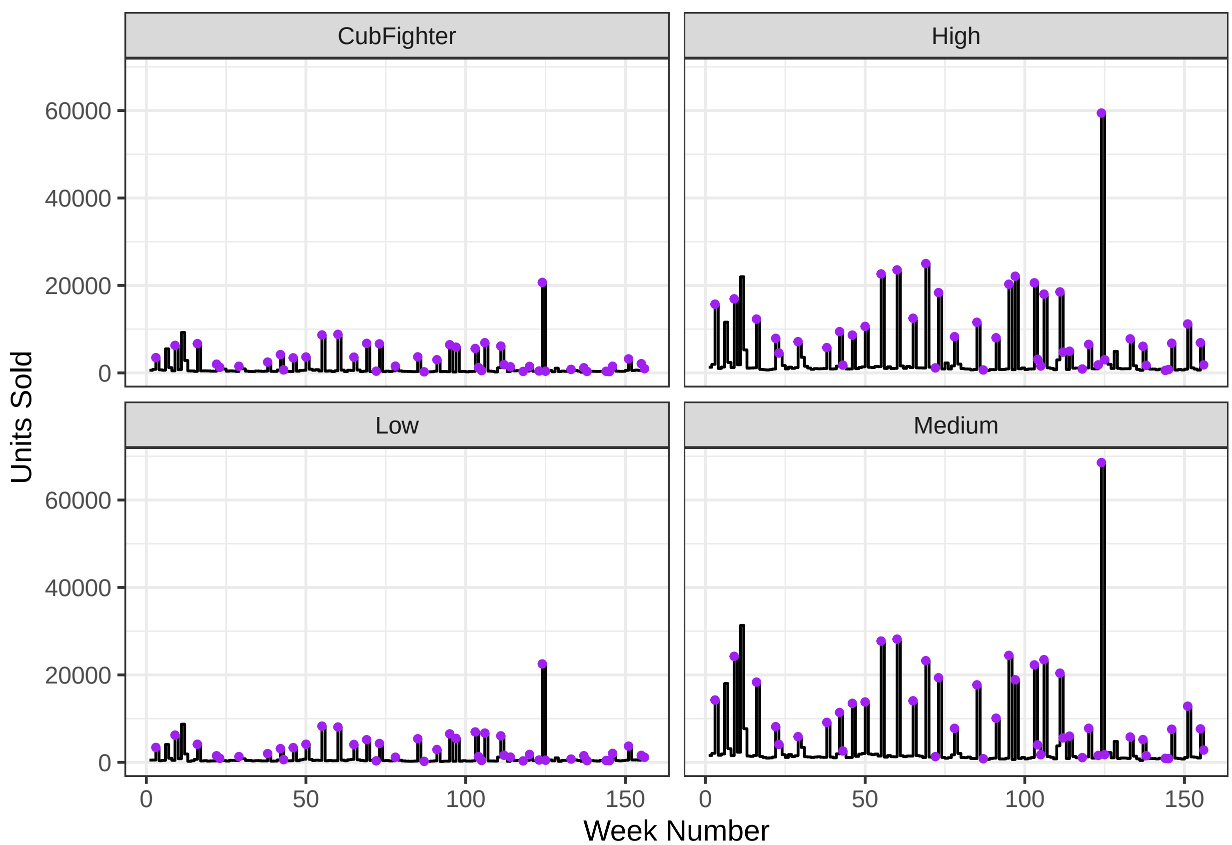
2.1 Data Exploration

2.1.1 Week / Year Breakdown

2016: Weeks 1-52 2017: Weeks 53-103 2018: Weeks 140-253

2.1.2 Exploratory Visualizations

```
# Plot units sold over time by zone
ggplot(df, aes(x = weeknum, y = units_tr_12)) +
  geom_step() +
  geom_point(data = df[df$merch_tr_12 == 1, ],
             aes(x = weeknum, y = units_tr_12),
             color = "purple",
             size = 1) +
  facet_wrap(~ zone, ncol = 2) +
  labs(x = "Week Number", y = "Units Sold")
```

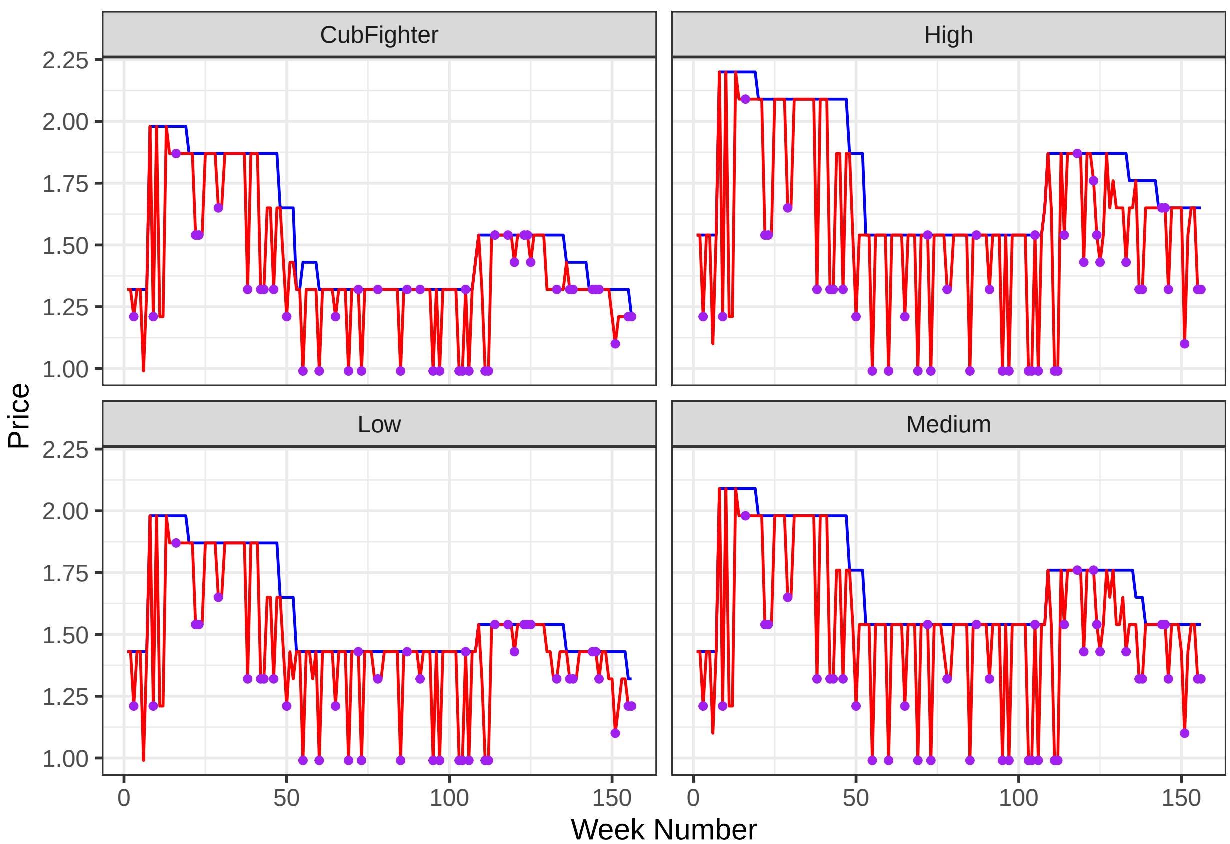


High and Medium zones consistently sold the more units than Low and Cubfighter zones.

Tropicana 12-ounce orange juice (Tr-12) sales is characterized by large spikes at more or less regular intervals between weeks 1 and 16 and between weeks 55 and 111. Tr-12 noticeably sold less units on all other weeks except for week 124. Sales were especially low in the week 17-42 period and the week 112-150 period (except for week 124).

Jewel implemented merchandising at almost all weeks where there was a spike in units sold.

```
# Plot prices over time by zone
ggplot(df, aes(x=weeknum)) +
  geom_line(aes(y=reg_tr_12), color="blue") +
  geom_line(aes(y=price_tr_12), color="red") +
  geom_point(data = df[df$merch_tr_12 == 1, ],
             aes(x = weeknum, y = price_tr_12),
             color = "purple",
             size = 1) +
  facet_wrap(~ zone, ncol=2) +
  labs(x="Week Number", y="Price")
```



Each zone hit its highest prices before week 50, dropped to a significantly lower base price with promotions dropping to $0.99 until week 111, slightly bumped up the price at about week 112, and then reduced the price again at about week 130 for the rest of the recorded time.

Prices in Medium and High zones reached a maximum of about $2.10 and $2.20, respectively. Prices in Low and Cubfighter zones never rose above $1.99. Each zone's price promotions dip down to $0.99 / unit at times.

Pricing trends seems to fit into three time periods, which may correspond with the years 2016, 2017, and 2018.

The promotional strategy in 2016 was relatively inconsistent. The strategy until about week 16 was to charge high prices with regular deep price promotions. This period corresponds with the initial high volumes sold from the previous graph. Then, the base prices decreased slightly, with promotions becoming shallower and more infrequent until week 38. Regular prices were significantly cut at the end of 2016, probably in an attempt to restore quantity sold to its early 2016 levels.

The promotional strategy for 2017 was to have a relatively low base price of $1.32 for the lower price zones (Cubfighter and Low) and $1.54 for the higher price zones (High and Medium), with regular price promotions dropping down to $0.99 for all four zones.

In early 2018, the promotional strategy was to raise base prices and reduce the frequency and magnitude of price promotions. Even when Tr-12's base price dropped again, there was only one relatively deep price promotion.

With the exception of a few weeks in 2016, merchandising was always implemented when promotional prices were offered.

Interestingly, there is no significantly lower price promotion at week 124 to justify the massive spike in units sold. Let's take a closer look at the week 124 quantity outlier.

2.1.3 Outlier Analysis

```r
# Calculate outlier cutoff for each zone
df <- df %>%
  group_by(zone) %>%
  mutate(
    q1 = quantile(units_tr_12, 0.25),
    q3 = quantile(units_tr_12, 0.75),
    iqr = IQR(units_tr_12),
    outlier_cutoff = round(q3 + 1.5 * iqr)
  ) %>%
  ungroup()

# Create a subset of the dataframe for only week 124 values.
subset = df[df$weeknum == 124, ]
q_wk_124 = subset$units_tr_12
subset[, c("zone", "units_tr_12", "outlier_cutoff")]
```

```
# A tibble: 4 x 3
```

```
  zone       units_tr_12 outlier_cutoff
  <chr>            <dbl>          <dbl>
1 CubFighter       20692           1804
2 High             59442           6130
3 Low              22510           1596
4 Medium           68563           6438
```

The quantity sold values in week 124 are significantly greater than the typical outlier cutoff (1.5 * IQR). Because these values are so disproportionately large with no intuitive rationale, I am going to remove them from the dataset.

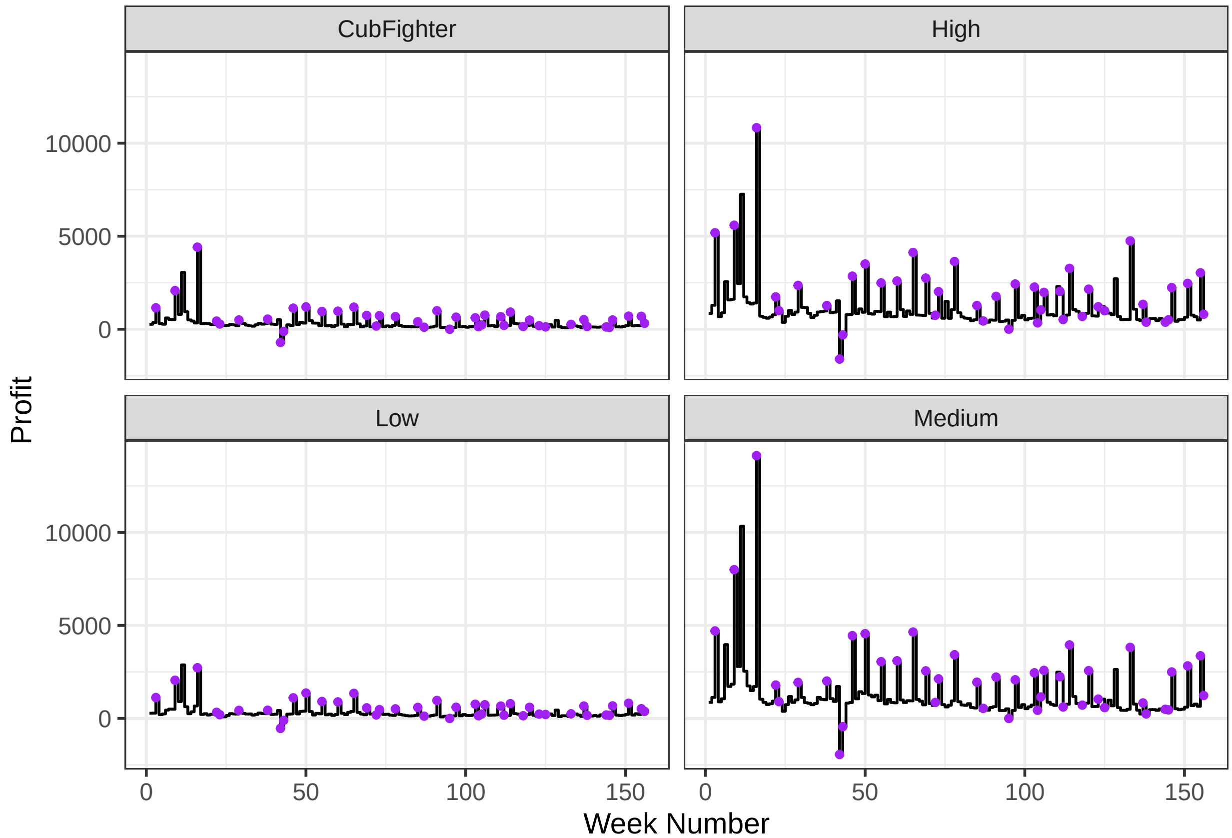
2.1.4 Deleting/Adding Columns

```r
# Remove week 124 values from the dataset
df = df[df$weeknum != 124,]
```

Next I am going to create a revenue column, calculated by multiplying the prices and the quantities.

2.1.5 Profit over Time

```r
# Create a revenue column and plot profit over time by zone
df$revenue_tr_12 = df$price_tr_12*df$units_tr_12
df$profit_tr_12 = (df$price_tr_12 - df$cost_tr_12) * df$units_tr_12
ggplot(df, aes(x = weeknum, y = profit_tr_12)) +
  geom_step() +
    geom_point(data = df[df$merch_tr_12 == 1, ],
             aes(x = weeknum, y = profit_tr_12),
             color = "purple",
             size = 1) +
  facet_wrap( ~ zone, ncol=2) +
  labs(x="Week Number", y="Profit")
```



The High and Medium zones make significantly higher profits than the Low and Cubfighter zones.

Merchandising was implemented at the beginning of both high and low profit spikes.

Profit spikes are the highest between weeks 9 and 16, corresponding to relatively higher prices, relatively frequent price promotions dipping slightly below $1.25, and high unit sales. Profits are negative for weeks 42 and 43 (toward the end of 2016), right before Jewel slashed its prices near the end of 2016.

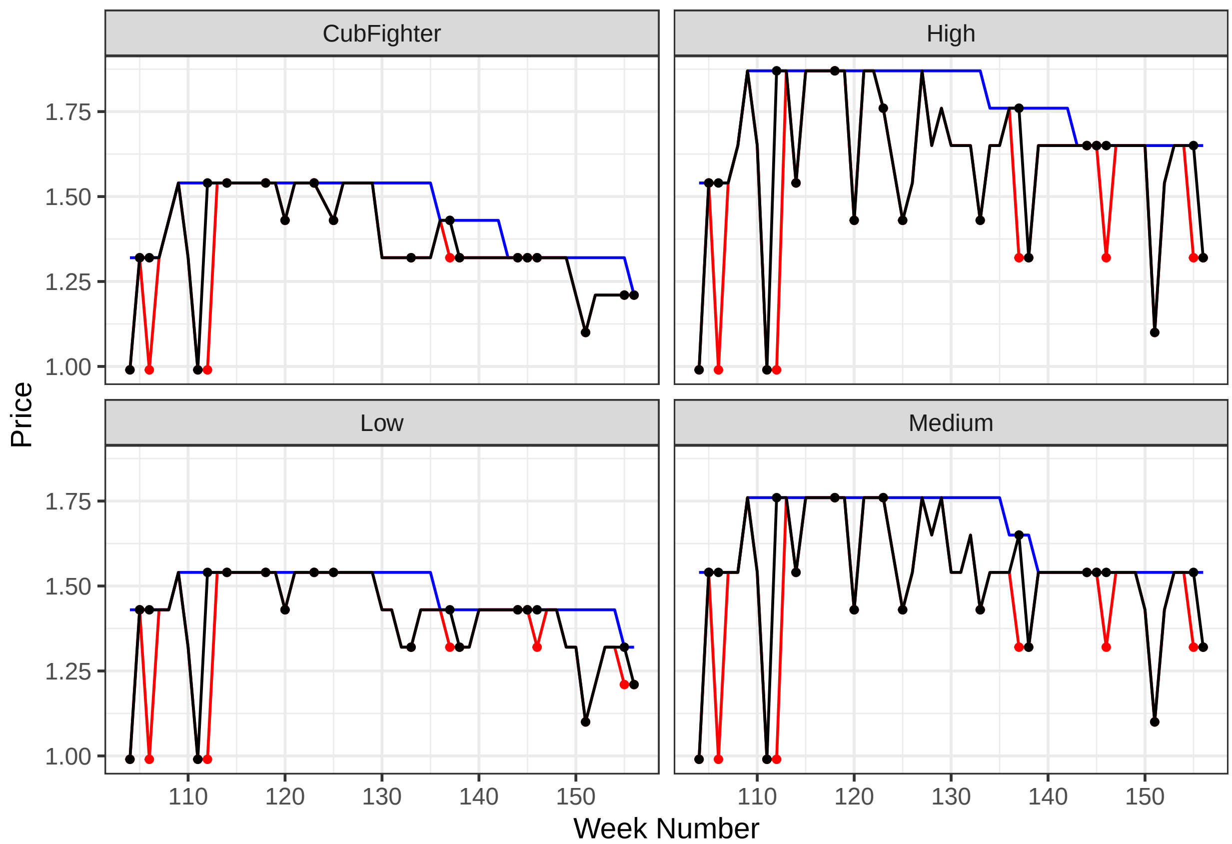
After the price cuts, profits rose again, fluctuating between the $4,000s and slightly below $1,000.

2.1.6 2018 Prices v. Scenario 1 Comparison

```
# Make a data frame for just 2018 data points
df_2018 = df[df$year == 2018, ]
```

```
# Plot a graph contrasting 2018 base, actual, and alternate scenario
↪ (Scenario 1) prices
ggplot(df_2018, aes(x=weeknum)) +
```

```r
geom_line(aes(y=reg_tr_12), color="blue") +
geom_line(aes(y=price_tr_12), color="red") +
geom_line(aes(y=price_tr_12_scenario1), color="black") +
geom_point(data = df_2018[df_2018$merch_tr_12 == 1, ],
           aes(x = weeknum, y = price_tr_12),
           color = "red",
           size = 1) +
geom_point(data = df_2018[df_2018$merch_tr_12 == 1, ],
           aes(x = weeknum, y = price_tr_12_scenario1),
           color = "black",
           size = 1) +
facet_wrap(~ zone, ncol=2) +
labs(x="Week Number", y="Price")
```



Scenario 1 features more price promotions than Jewel offered in 2018, especially in the High and Medium zones. With this alternate strategy, Jewel would be mostly relying on the strategy that worked in early 2016 and 2017 – offering price promotions every few weeks to boost quantity sold, especially in the high-price and high-quantity High and Medium zones.

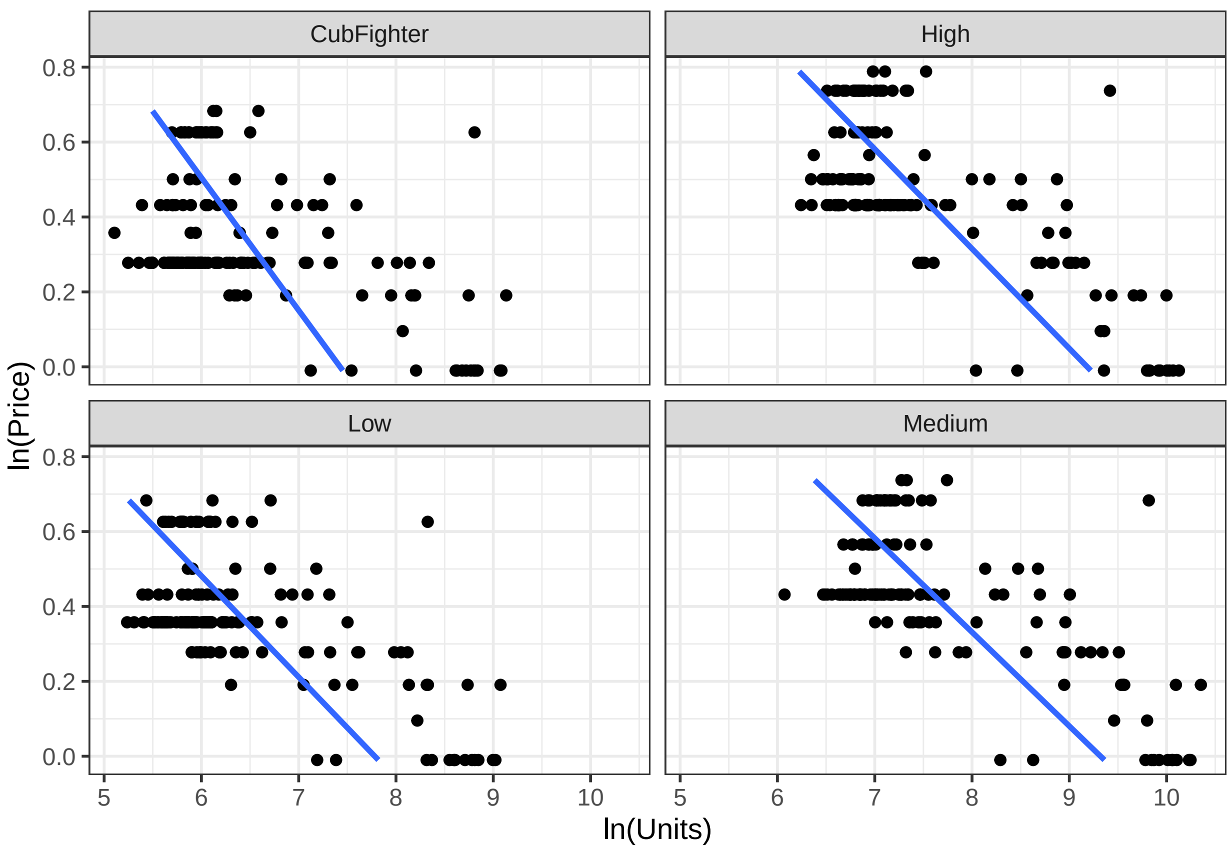
Scenario 1 also incorporates more merchandising than was offered in 2018, offering it at almost every price promotion, as well as some weeks where price promotions are not offered.

While scenario 1 is in some ways a return to pre-2018 pricing strategy, it offers a sort of middle ground between the 2017 and 2018 promotional strategies. While scenario 1 has more frequent and deeper price promotions than Jewel offered in 2018, base and promotional prices are still higher than they were in 2017, allowing them to capture a greater profit per unit than in 2017.

2.1.7 Price - Units Sold Relationship

Finally, let's compare ln_p and ln_q to analyze the correlation between price and units sold.

```
# Plot Log Price vs. Log Units
ggplot(df, aes(x=ln_q, y=ln_p)) +
  geom_point() +
  facet_wrap( ~ zone, ncol=2) +
  geom_smooth(method='lm', se=FALSE, orientation="y") +
  labs(x="ln(Units)", y="ln(Price)")
```



Across all four zones, there is a clear negative correlation between price and the number of units sold. This means that more units are typically sold when prices are lower.

Next, let's run a regression model for Tr-12 data.

2.2 Main Regression Model

```
reg.tr <- lm(ln_q ~ ln_p + ln_p:Dmerch + Dmerch +
               Dyear + Dzone, data=df)
summary(reg.tr)
```

```
Call:
lm(formula = ln_q ~ ln_p + ln_p:Dmerch + Dmerch + Dyear + Dzone,
    data = df)

Residuals:
     Min       1Q   Median       3Q      Max
-1.76713 -0.27890 -0.01948  0.27835  2.37613

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   7.68458    0.10992  69.912  < 2e-16 ***
ln_p         -2.79752    0.20211 -13.841  < 2e-16 ***
Dmerch1       1.39015    0.10845  12.818  < 2e-16 ***
Dyear2017    -0.85938    0.06077 -14.140  < 2e-16 ***
Dyear2018    -0.85055    0.05559 -15.301  < 2e-16 ***
DzoneHigh     1.34960    0.06327  21.332  < 2e-16 ***
DzoneLow      0.04112    0.05951   0.691     0.49
DzoneMedium   1.41270    0.06162  22.925  < 2e-16 ***
ln_p:Dmerch1 -1.42485    0.27983  -5.092 4.73e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5215 on 611 degrees of freedom
Multiple R-squared:  0.798, Adjusted R-squared:  0.7953
F-statistic: 301.6 on 8 and 611 DF,  p-value: < 2.2e-16
```

```
elasticity = -2.80
merch_eff = round(exp(1.39015), 2)
merch_elasticity_eff = -1.42
elasticity_merch = elasticity * merch_elasticity_eff

paste("Baseline price elasticity: ", elasticity)
```

```
[1] "Baseline price elasticity:  -2.8"
```

```r
paste("Baseline merchandising multiplier: ", merch_eff)
```

```
[1] "Baseline merchandising multiplier:  4.02"
```

```r
paste("Elasticity with merch effect: ", elasticity_merch)
```

```
[1] "Elasticity with merch effect:  3.976"
```

The price elasticity of Tr-12 is -2.8 , holding all else constant. This means that consumers are slightly more price sensitive to Tr-12 prices than they would be for the average product.

During merchandising weeks, Jewel sells 4.02 times as much compared to non-merchandising weeks, holding all else constant.

During merchandising weeks, the price elasticity is 3.976. This means that consumers are more responsive to price changes during merchandising weeks. As a result, they are more likely to increase consumption when prices are promoted if merchandising occurs at the same time as promotion. Based on the exploratory data analysis above, Jewel seems to be employing this tactic well.

2.3 Scenario Analysis

```r
# Create a copy of the original df for Scenario 1 predictions
df.pred = df
```

2.3.1 Scenario 0 (Baseline)

```r
# Predict ln(units) using the actual data -- the output are the fitted
↪  values
reg.tr <- lm(ln_q ~ ln_p + ln_p:Dmerch + Dmerch +
               Dyear + Dzone, data=df.pred)
df.pred$ln_pred_units_tr_12_0 = predict(reg.tr, df.pred)

# Convert Log to Units and Compute Profit
df.pred = df.pred %>%
  mutate(pred_units_tr_12_0 = exp(ln_pred_units_tr_12_0),
         profit_tr_12_0 = pred_units_tr_12_0 * (price_tr_12 - cost_tr_12))

# Create a table to summarize predicted profit for baseline
```

```r
df.pred %>%
  group_by(year) %>%
  summarize(sum_profits_0 = round(sum(profit_tr_12_0))) %>%
  kable()
```

year	sum_profits_0
2016	184272
2017	119693
2018	121653

2.3.2 Scenario 1

```r
# Implement the scenario!
# This is *not* the same scenario from the Energy Bar case in class.
# You need to modify the original data to alter the price and merchandise
↪  variables
# using the *_scenario variables that we provide.

# Use the regression model to predict using the modified data
df_scenario1 = df
df_scenario1$ln_p = log(df$price_tr_12_scenario1)
df_scenario1$Dmerch = factor(df$merch_tr_12_scenario1)

df_scenario1$ln_pred_units_tr_12_1 = predict(reg.tr, newdata =
↪  df_scenario1)


# Convert Log to Units and Compute Profit
df_scenario1 = df_scenario1 %>%
  mutate(pred_units_tr_12_1 = exp(ln_pred_units_tr_12_1),
         profit_tr_12_1 = pred_units_tr_12_1 * (price_tr_12_scenario1 -
          ↪  cost_tr_12))

# Create a table to summarize predicted profit for the alternative
↪  scenario
df_scenario1 %>%
  group_by(year) %>%
  summarize(sum_profits_1 = round(sum(profit_tr_12_1))) %>%
  kable()
```

year	sum_profits_1
2016	NA
2017	NA
2018	104342

```
 difference = 121653-104342
difference
```

```
[1] 17311
```

The predicted profit for 2018 from scenario 1 (more frequent and deeper promotions than scenario 0) is $17,311 less than the predicted profit from scenario 0 ($104,342 v. $121,653). This means that Jewel may be better off offering less frequent and shallower price promotions.

2.4 Takeaways

2017 was a tough year for Jewel because they tried to boost unit sales at the expense of profit per unit to make up for the negative profits incurred near the end of 2016. They tried to correct for that in 2018 by minimizing price promotion frequency and magnitude.

However, the 2018 scenario analysis indicates that Jewel would benefit from offering more frequent and slightly deeper price promotions, offering a sort of middle ground between what they did in 2017 and what they did in 2018. A new promotion strategy based on scenario 1 would decrease Jewel profits because it would cuts too deeply into profit margins, taking away money that Jewel could otherwise have.